

Modelling Insurgencies through Agent-Based Simulations

Professor-ready research memo | current gate state as of 2026-09-08

Decision

The computational program now has a reproducible exactness and synthetic-inference spine, but it is not authorized to promote a historical transfer claim. The current packet keeps the historical gate closed because the live core differs from the prior certificate and the fixed production throughput gate must be independently recorded before historical confrontation.

What is established

Gate	Evidence	Status
Exact scheduler oracle	12 seeds x 5 horizons; world and continuation hashes	PASS
Packed authority	Runtime schema inventory with explicit sparse boundaries	PASS
Kernel oracles	Physical and numeric information kernel comparisons	PASS
Synthetic inference	RB, guided, MCSE, combined validation; no history	PASS

Scientific interpretation

The model is treated as a latent-state mechanism: organizational capacity, civilian access, territorial control, belief state, logistics readiness, operational posture, and local exposure jointly determine action hazards and foothold reproduction. The central quantity is R_t , the expected number of newly active footholds produced per active foothold over a defined interval. This is a falsifiable mechanism statement, not an assertion that historical fit has been achieved.

Current blockers

Blocker	Consequence
Core freeze drift	Prior v5 historical artifacts are stale-core evidence and cannot be promoted.
Performance gate	The complete fixed 768-PWB run exceeded the safe resource window and was stopped; no PWB/s value is reported and E1/E2 remain failed closed.
Historical confrontation	Nepal and Afghanistan are not rerun unless the frozen method contract authorizes the once-only run.

Recommended next decision

Treat this packet as a methods and audit handoff. If the throughput gate passes and the core boundary is frozen, run the preregistered Nepal and Afghanistan confrontation exactly once, preserve every seed and negative result, and report calibration-free predictive scores against predeclared observation operators. If the gate fails, optimize the identified packed synchronization bottleneck before any historical rerun.

Do not claim

Claim	Why it is withheld
The model explains Nepal or Afghanistan	No authorized current-core historical confrontation is present in this packet.
The mechanism is identified historically	Synthetic identification is passed; historical identification is not established.
The model is production-ready	Throughput and freeze conditions must be evaluated from the current fixed-work artifact.

Reproducibility

All evidence is content-hashed and tied to the current model hash, source state, and explicit protocol. The negative guided-validation v1 artifact is preserved alongside the passing v2 revision; no failed run is deleted or silently reclassified. No outreach or external communication is performed by this packet builder.

Packet contents

Architecture figure; technical note; fixed-work performance result; claims/evidence matrix; limitations and failure ledger; machine-readable manifest; and this readme. The accompanying JSON manifest is the authoritative file inventory.